WESTERN PACIFIC UNIVERSITY

CMN224 / CMP224 Software Engineering — Assessment

Task 2: Requirements & Estimation Report

Client brief — released Week 2 · due end of Week 8 (Moodle submission) · 15% of final grade.

Unit Coordinator: Mr Solomon Hombiku



Task 2: Requirements & Estimation Report

INTRODUCTION

This report sets out the requirements and delivery plan for a client system chosen from the scenario; Kina Ridge Primary School fee-tracking system. As the commissioned software engineer, the task is not to build the solution immediately but to establish what the client needs and to provide an honest assessment of how long delivery will take and in what order. The report is structured in two parts: a concise Software Requirements Specification (SRS) and a realistic estimate with work schedule.

OBJECTIVE

The objective of this commission is to:

- Identify client needs by analyzing their business context, stakeholders, and operational environment.
- Define system requirements through functional, non-functional, constraints, and business rules, ensuring clarity on what is in and out of scope.
- Provide delivery estimates by breaking down tasks, assigning timeframes, and setting milestones that the client can track.
- Offer dependency-triangle analysis to explain trade-offs between scope, time, and cost, giving the client a transparent basis for decision-making.

For the Kina Ridge Primary School fee system in Port Moresby, the software needs a lightweight, offline-first desktop app with local sync, automated paper-receipt printing, and SMS/digital payment logging to handle weekly blackouts, cash/mobile money tracking, and arrears reporting for 450 students.

PART 1 — CONCISE SRS (TAUGHT WEEK02)

1.1 Purpose & Scope

- **Purpose:** Track school fee payments, generate reliable paper receipts, and show unpaid fee balances clearly. Provide a reliable fee-tracking system for Kina Ridge Primary School to reduce arrears, ensure receipts are trustworthy, and generate clear reports.
- **In Scope:** Student records, payment logging (cash, bank, mobile money), receipt printing, arrears reports, and role-based views. It also covers recording of payments (cash, bank deposit, mobile money), issuing receipts, tracking arrears, and producing summaries for staff and board.
- **Out of Scope:** Online parent payment portals, full school accounting (payroll/ledger), and mobile apps for parents. Also covers Payroll, academic

records, student performance, external integrations beyond basic payment recording.

1.2 Business Context & Stakeholders

- **Bursar**: Needs fast payment entry and daily reconciliation. Also needs simple entry of payments and automatic receipt generation.
- **Office Assistants**: Need simple data entry for cash and receipts. And need to help with data entry and printing receipts.
- **Head Teacher**: Needs instant access to student fee status and summaries. Needs arrears reports before exams.
- **Board**: Needs termly arrears reports and financial overviews. Needs term summaries of fee collection and arrears also as well.

1.3 Functional Requirements (MoSCoW-prioritised)

- **FR-01 (Must)**: The system shall record fee payments split across cash, bank deposits, and mobile money instalments.
- **FR-02 (Must)**: The system shall print a physical receipt for every logged payment using the office printer.
- **FR-03 (Must)**: The system shall calculate and display total outstanding arrears per student and grade level.
- **FR-04 (Should)**: The system shall generate termly summary reports for the head teacher and board.
- **FR-05 (Could)**: The system shall export financial summaries to Excel for legacy backup.
- **FR-06 (Wont)**: The system shall not process online credit card transactions in version 1.

NB: Pattern: "The system shall <verb> <object> <qualifiers>" - one behaviour per statement, each one testable.

1.4 Non-Functional Requirements

- **Reliability:** The system shall store all transactions in a local database with automatic save intervals of 2 minutes, ensuring zero data loss during weekly power cuts common in Port Moresby.

- **Usability:** Core payment entry (student ID, amount, method, date) shall be completed by office staff in 60 seconds per transaction after 2 hours of training, reflecting the limited digital literacy and resource constraints in PNG schools.
- **Performance:** Arrears reports covering all 450 students shall load in 3 seconds on the school's existing Windows laptop, even when operating offline during Wi-Fi outages.
- **Security:** Role-based access shall restrict fee editing rights to the bursar only, while the head teacher and board members shall have read-only access. This ensures compliance with school governance practices in PNG, where financial authority is centralized.

1.5 Constraints & Business Rules

- **Hardware Limit**: Must run on a single shared office laptop and existing printer.
- **Power & Net**: Must operate smoothly during weekly power outages and intermittent Wi-Fi.
  Constraints: One Windows laptop, one printer, weekly power cuts, intermittent Wi-Fi.
- **Business Rule**: Partial instalment payments are fully permitted and must accumulate toward the total fee balance.
- Receipts must be sequentially numbered.
- Students with <50% fees paid cannot sit exams.
- Reports must be produced at the end of each term.

1.6 Glossary

- **Bursar**: The staff member who handles money and school finances.
- **Arrears**: Money that is owed and past the payment deadline.
- **MoSCoW**: A way to rank features by priority (Must, Should, Could, Won't).
- **Receipt:** Printed proof of payment with unique number.
- **Reliability:** Means the system consistently performs its functions without failure.
- **Usability:** Is how easy and efficient the system is for staff to operate.
- **Performance:** Measures how quickly and effectively the system responds.
- **Security:** Ensures that only authorized users can access or modify sensitive information.

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PART 2 — ESTIMATE & WORK SCHEDULE (TAUGHT WEEKS 5–6)

2.1 Task breakdown

- Requirements validation
- Database design
- Payment entry module.
- Receipt Generation.
- Arrears flagging.
- Reporting Module.
- Testing
- Training & Deployment

Task Breakdown Table

Task	Description
Requirements validation	Meet bursar, assistants, head teacher, board to confirm needs
Database design	Create student + payment tables with arrears logic
Payment entry module	Interface for recording cash, bank, mobile money payments
Receipt generation	Auto-numbered receipts printable on existing printer
Arrears flagging	Logic to highlight unpaid students before exams
Reporting module	Summaries for head teacher and board each term
Testing	Run the system with sample student and payment data to check that each module works correctly (payment entry, receipts, arrears flagging, reporting).
Training & deployment	Train staff (≤2 hrs) and install on office laptop

NB: Table 1.0. Displaying Estimate and Work Schedule of the Task Breakdown.

2.2 Time estimates (assumptions: 1 developer, 30 hrs/week)

- Requirements validation: 1 week.
- Database + Payment Module: 2 weeks.
- Receipt Generation: 1 week.

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- Arrears flagging: 1 week.
- Reporting module: 1 week.
- Testing: 1 week.
- Training + deployment: 1 week.
  Total: 8 weeks.

- **Time Estimates Table**

Task	Estimate	Assumptions & Reasoning
Requirements validation	1 Week = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Meetings + clarifications with stakeholders
Database + Payment Module	2 Weeks = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Core structure; extra time for offline resilience
Receipt Generation	1 Week = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Simple print integration with existing hardware
Arrears flagging	1 Week = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Straightforward logic; tested against 450 records
Reporting module	1 Week = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Summaries; accounts for term cycles
Testing	1 Week = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Includes power cut/Wi-Fi outage simulations
Training + deployment	1 Week = 5 workdays/week (8hrs/day * 5 = 40 hours/week).	Staff training (≤8 hrs) + installation
Total	280 hours on 8 weeks	1 developer, 40 hrs/week, PNG delivery conditions considered

NB: Table 1.1. Showing the estimate and work schedule for Time Estimate Table.

2.3 Work schedule

- **Milestone 1 (Week 1):** Requirements Validate
- **Milestone 2 (Week 3):** Payment entry + Receipt
- **Milestone 3 (Week 5):** Arrears flagging
- **Milestone 4 (Week 6):** Reporting module
- **Final delivery (Week 8):** Final delivery

Work Schedule Table

Week	Milestone	Client Deliverable
Week 1	Requirements validated	Confirmed scope & stakeholder needs
Week 3	Payment entry + receipts	Staff can record payments and print receipts
Week 5	Arrears flagging	Head teacher sees arrears before exams
Week 6	Reporting module	Board receives term summaries
Week 8	Final delivery	Tested, trained, deployed system

NB: Table 1.2. Demonstrating the estimate and work schedule for the Work Schedule Table.

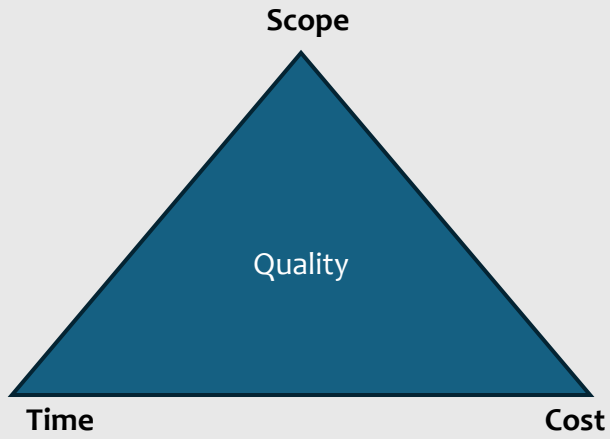
2.4 Dependency-triangle analysis

- **Scope:** Core functions (payment, receipts, arrears) are non-negotiable.
- **Time:** 8 weeks realistic; rushing risks errors, 5 workdays/week (8hrs/day * 5 = 40 hours/week).
- **Cost:** One developer, existing hardware.

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- **Recommendation:** If the board demands faster delivery, reduce scope (e.g., delay reporting module) rather than compromise reliability.

- ### The Dependency triangle



NB: The Drawn Image of Dependency Triangle.

The Dependency Triangle Analysis Table

Corner	Trade-off	Recommendation
Scope	Core features (payments, receipts, arrears) non-negotiable; reporting can be delayed if needed	Protect reliability by reducing scope if time is cut
Time	8 weeks realistic; rushing risks errors and poor training	Stick to 8 weeks unless scope reduced
Cost	One developer, existing laptop/printer; no extra licenses	Keep cost low by using current hardware

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Overall	Honest plan reflects PNG constraints (power cuts, Wi-Fi outages, limited staff capacity)	If board demands faster delivery, drop reporting module first

NB: Table 1.3. Displaying the Estimate and work schedule for Dependency Triangle Analysis Table

CONCLUSION

This report has established the requirements and delivery plan for the chosen client scenario, ensuring that stakeholder needs, PNG delivery constraints, and realistic estimates are addressed. The proposed system focuses on reliable fee tracking, trustworthy receipts, and clear arrears reporting, with milestones designed to deliver visible value early. The dependency-triangle analysis highlights that scope, time, and cost are interdependent, and the most sustainable trade-off is to reduce scope (e.g., delay reporting features) rather than compromise reliability or staff training. Overall, the plan provides the client with a transparent, defensible roadmap for successful system implementation.

REFERENCES

- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications. Available via Google search: IEEE Std 830-1998 SRS (google.com in Bing)
- Course Workshop Notes, Week 2 (Concise SRS techniques).
- Course Workshop Notes, Weeks 5–6 (Estimation and scheduling methods).
- Client scenario narratives: Kina Ridge Primary School and Waghi Valley Produce Cooperative (provided in commission brief).

AI Use: I used AI support because it helps me take complex or lengthy information and turn it into shorter, clearer, and more understandable points. Instead of manually condensing large amounts of detail, AI can quickly highlight the most important ideas, organize them logically, and present them in a way that is easy to grasp. This makes studying, preparing reports, or drafting assignments more efficient, while ensuring that the content remains accurate and focused on the key requirements.